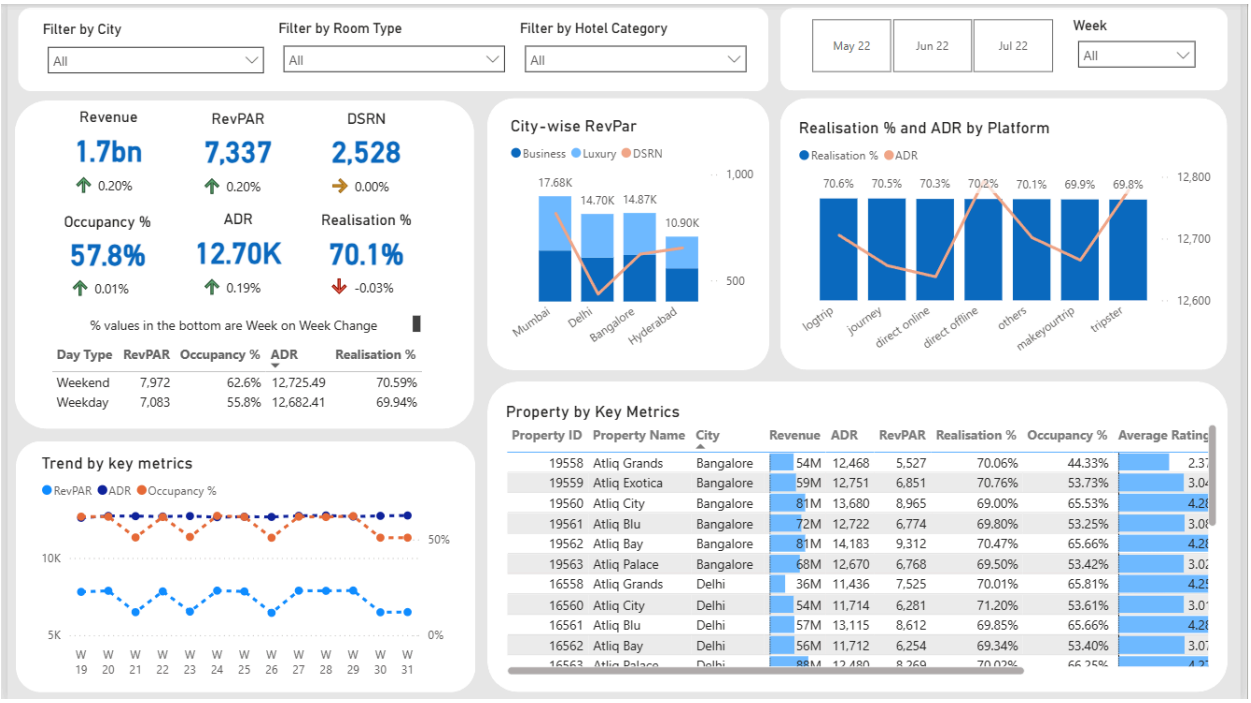
# Extracting revenue insights for a hospitality company

For this project, I am using dummy data from a hospitality company to build a Power BI dashboard and gather insights on which hotels are performing well and which strategies can be implemented to improve the business.




## Understanding the data

The data consisted of 5 corresponding tables:

1. **dim_date**: Acts as the time dimension for bookings, enabling consistent filtering, grouping, and time-based analysis (daily, weekly trends).
2. **dim_rooms**: Types of rooms provided by the hotel company.
3. **dim_hotels**: Contains hotel metadata (property_id, city, category, property)
4. **fact_aggregated_bookings**: Date-wise summarization of successful bookings and room capacity of hotels
5. **fact_bookings**: Captures granular booking-level data (dates, guests, revenue, status), forming the core transactional dataset for analyzing performance and customer behavior.

# Transforming the data

Most of the data didn't need any transformation except the day_type column in the dim_date table. According to the hospitality industry, Friday and Saturday are considered as weekends rather than Saturday and Sunday. Therefore, the day_type column was transformed using the code below:

```
day_type =
var wkd = WEEKDAY(dim_date[date])
return if(wkd>5, "Weekend", "Weekday")
```

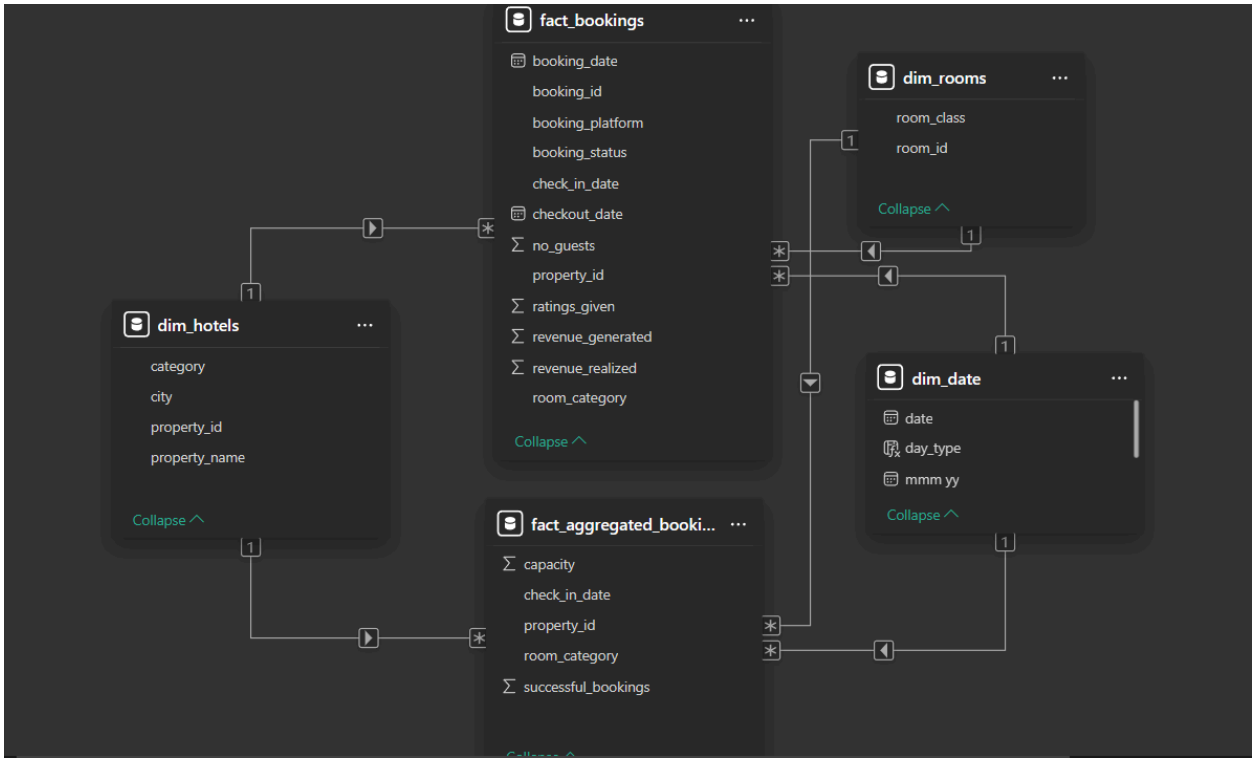
# Key Measures

I declared a total of 26 key measures to gain insights from the data, out of which the following metrics gave the major insights:

1. **ADR**: It is the ratio of revenue to the total rooms booked/sold. It is the measure of the average paid for rooms sold in a given time period
2. **RevPAR**: RevPAR represents the revenue generated per available room, whether or not they are occupied. RevPAR helps hotels measure their revenue-generating performance to accurately price rooms. RevPAR can help hotels measure themselves against other properties or brands.
3. **Realisation %**: The successful "checked out" percentage over all bookings happened.
4. **Occupancy %**: total successful bookings happened to the total rooms available(capacity)
5. **DSRN**: This metric shows the average number of rooms that are ready to sell for a day, considering a time period

# Data Modelling

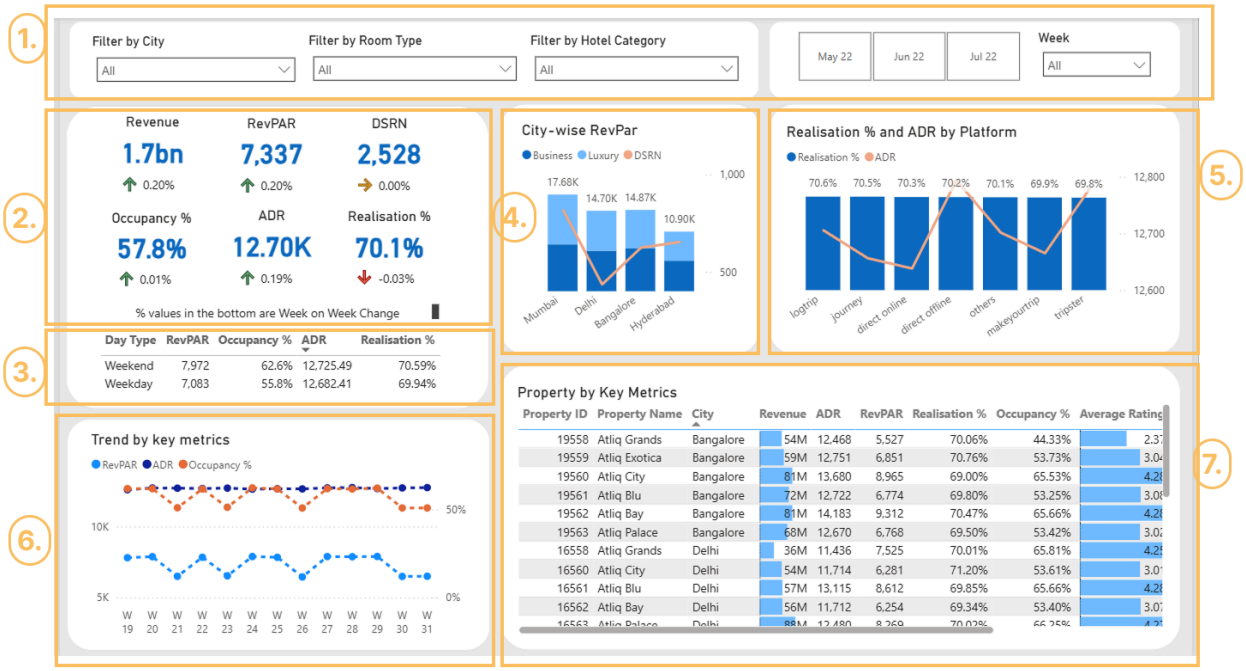
The following relationships were made in Model View:




# Exploratory questions

- Which city maximizes revenue per room?
- Which platform is most reliable?
- Where is pricing power strongest?
- Which properties are underutilized?
- Is weekend pricing optimized?

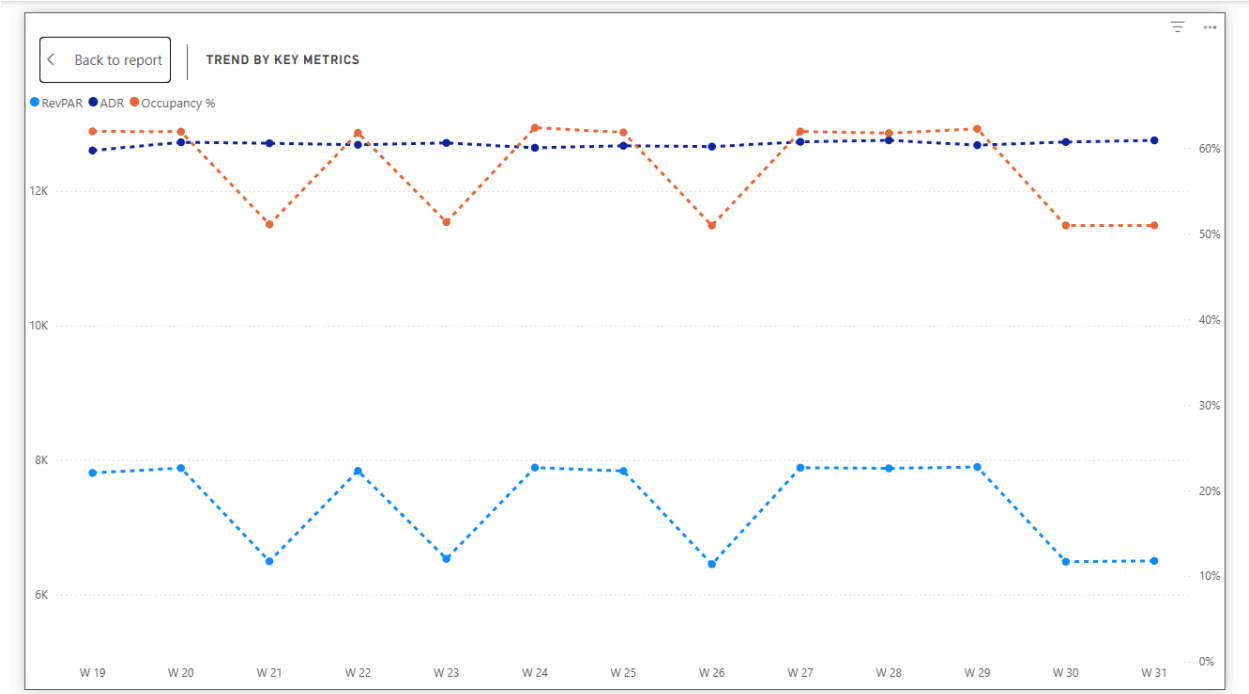
# Dashboard design




1. **Filters (City, Room Type, Category, Date, Week)**: Let users slice the data easily to compare performance across different segments and time periods.
2. **Top KPIs (Revenue, RevPAR, DSRN, Occupancy, ADR, Realisation)**: Give a quick snapshot of overall business performance and highlight week-on-week changes.
3. **Day Type Split (Weekend vs Weekday)**: Helps identify how demand and pricing differ between weekdays and weekends.
4. **City-wise RevPAR Chart**: Shows which cities are driving revenue efficiency and how business vs luxury segments perform across locations.
5. **Realisation % and ADR by Platform**: Helps understand which booking platforms are more profitable and efficient in pricing.
6. **Trend by Key Metrics**: Tracks how core metrics change over time, making it easier to spot upward or downward trends.
7. **Property by Key Metrics Table**: Enables detailed comparison of individual properties to identify top and underperforming hotels.

# Gathering insights

## Pricing strategy




ADR stays almost the same through May, June, and July, with a little fluctuation in the Occupancy percentage. This implies that the company is using a fixed price for their hotels throughout these 3 months.

| Day Type | RevPAR | Occupancy % | ADR | Realisation % |
|---|---|---|---|---|
| Weekend | 7,972 | 62.6% | 12,725.49 | 70.59% |
| Weekday | 7,083 | 55.8% | 12,682.41 | 69.94% |

The weekend and weekday aggregate also show a very slight difference in ADR and Occupancy%.

Therefore, we conclude that the hotel is not using any dynamic pricing strategy, and this is an opportunity to double down on. May, June and July are holiday seasons, and people travel a lot during this time. This company can segregate the hotels based on their Occupancy %, Average Ratings and RevPar, and place a higher price on the rooms.
The same can be planned for weekdays and weekends, where a higher price can be placed during the weekends.

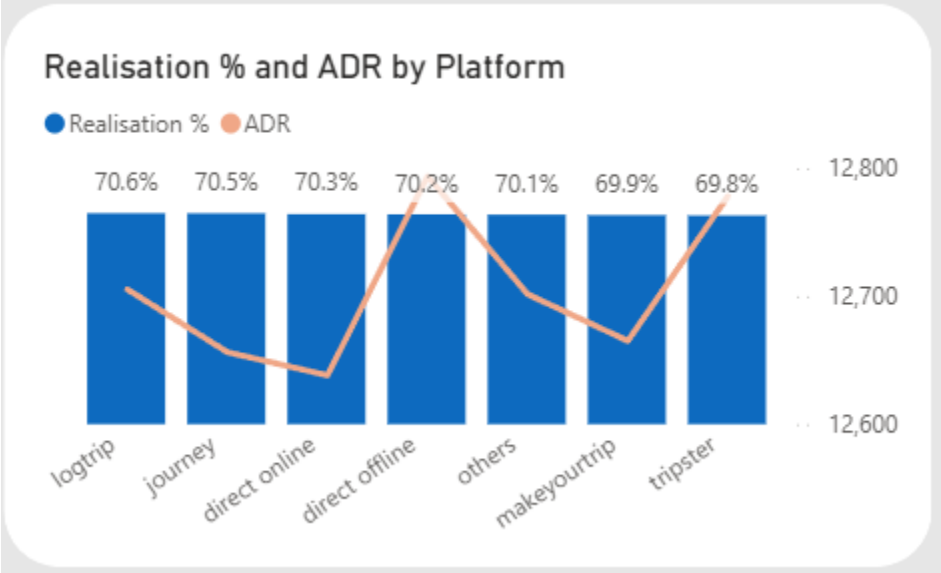
## Patterns in Occupancy %

Property by Key Metrics

| perty ID | Property Name | City | Revenue | ADR | RevPAR | Realisation % | Occupancy % | Average Rating | Ca |
|---|---|---|---|---|---|---|---|---|---|
| 19558 | Atliq Grands | Bangalore | 54M | 12,468 | 5,527 | 70.06% | 44.33% | 2.37 | |
| 17564 | Atliq Seasons | Mumbai | 65M | 16,597 | 7,397 | 70.59% | 44.57% | 2.30 | |
| 18559 | Atliq Exotica | Hyderabad | 47M | 9,111 | 4,061 | 70.57% | 44.57% | 2.33 | |
| 17562 | Atliq Bay | Mumbai | 51M | 15,167 | 6,803 | 69.60% | 44.86% | 2.37 | |
| 18563 | Atliq Palace | Hyderabad | 44M | 9,480 | 5,014 | 69.57% | 52.89% | 3.07 | |
| 17560 | Atliq City | Mumbai | 87M | 14,629 | 7,763 | 69.51% | 53.07% | 3.04 | |
| 19561 | Atliq Blu | Bangalore | 72M | 12,722 | 6,774 | 69.80% | 53.25% | 3.08 | |
| 18558 | Atliq Grands | Hyderabad | 46M | 10,331 | 5,514 | 69.73% | 53.38% | 3.06 | |
| 16562 | Atliq Bay | Delhi | 56M | 11,712 | 6,254 | 69.34% | 53.40% | 3.07 | |
| 19563 | Atliq Palace | Bangalore | 68M | 12,670 | 6,768 | 69.50% | 53.42% | 3.02 | |
| 17558 | Atliq Grands | Mumbai | 74M | 14,830 | 7,053 | 69.01% | 53.60% | 3.05 | |

When we arrange the table in the increasing order of Occupancy %, we discover an almost linear relationship with Average Rating for the hotels. The reviews and ratings of the hotels can be leveraged to improve the quality of rooms and hotel services, or we can lower the prices of such rooms/hotels, therefore leading to higher occupancy and a reasonable revenue.

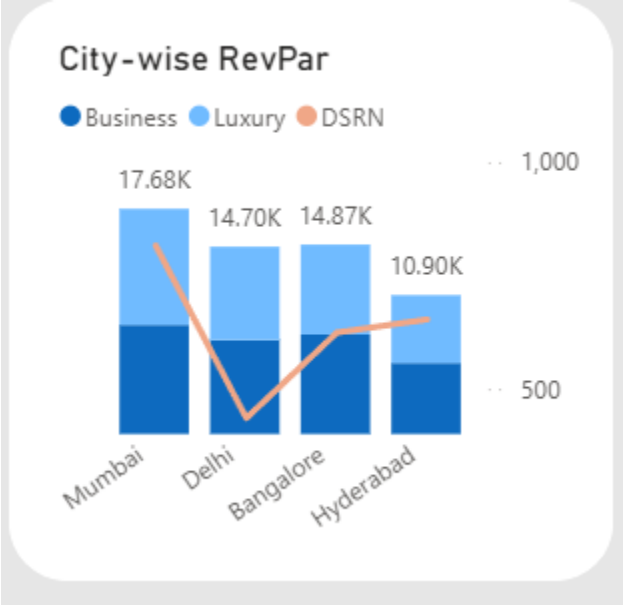
## Platform-wise data



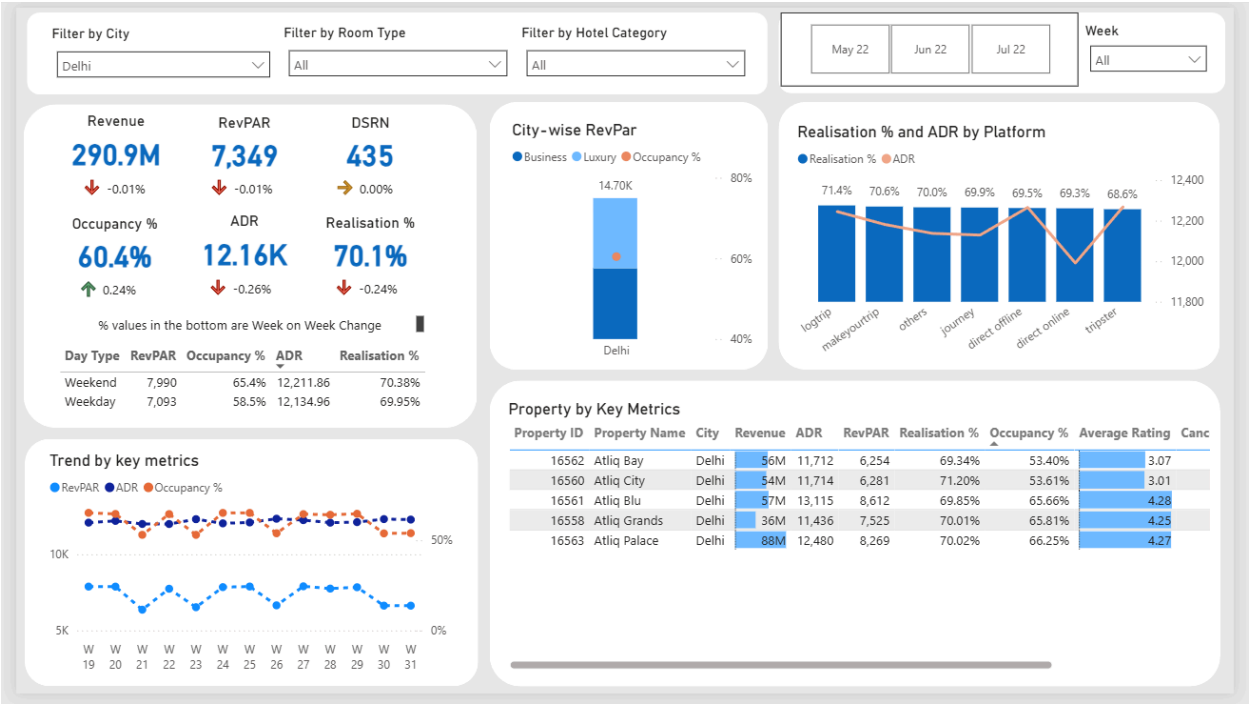

Most of the bookings are through the direct offline medium, whereas makeyourtrip, direct online and journey contribute to a very small percentage of successful checkouts at the hotel. The hotel listings on makeyourtrip and journey should be examined to find if there are any

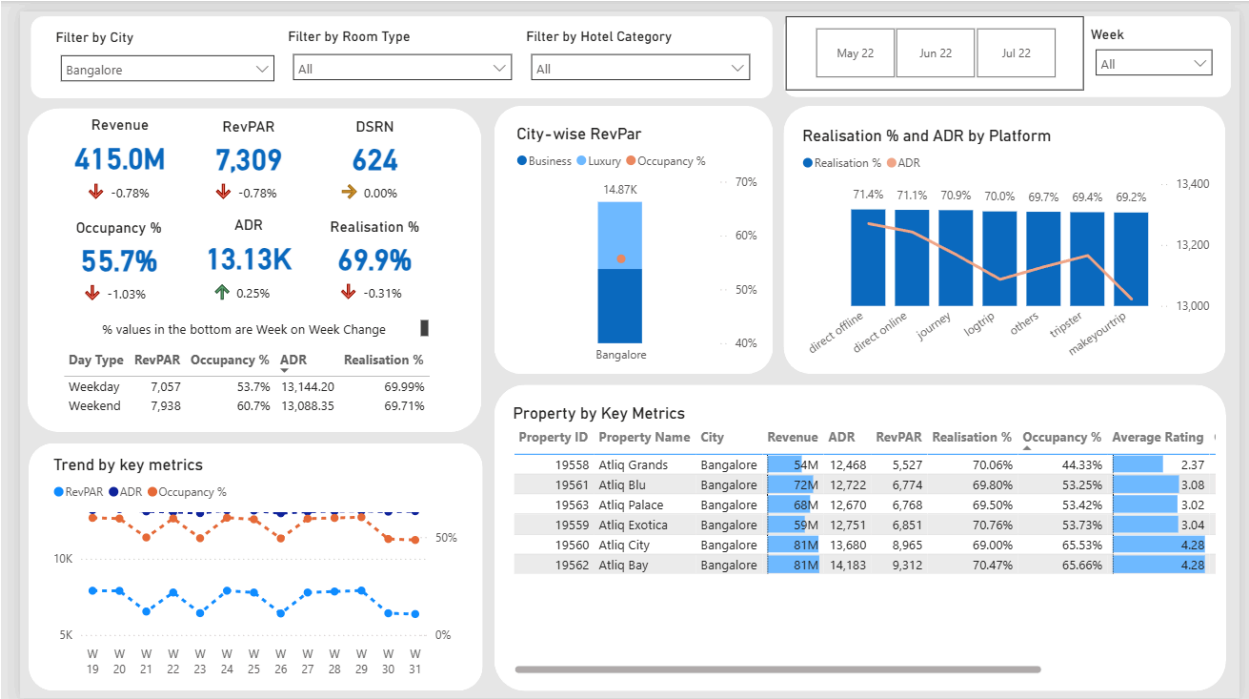
mismatches people face or if there is a pricing issue, and thus information should be updated accordingly.
Promotions can be done via the direct online medium i.e., the company's hotel booking website, to attract more customers. Since the company owes no commission to the third party using this medium, offering coupons and complementary services to people can be a good strategy.

## City-wise comparison: Delhi vs Bangalore




Both cities have an almost equal RevPar but a little difference in DSRN leads to 200 million more in revenue in Bangalore.




The RevPAR and Realization % for both cities are almost the same, indicating a strong performance in both regions but since there are more hotels present in Bangalore, the revenue realised is higher. A strategy to open more hotels in Delhi can be a great strategy to expand the company and generate more revenue, since it is as already working market.